# Daily Drilling Report

**Well ID: Forge 78B-32**
Field: UTAHFORGE

## Geothermal Resource Group, Inc.

**Well Name: 78B-32**
Sect: 32 Town: 26 Rng: 9W County: Beaver State: UT

**Report No:** 2 **Report For 28-Jun-21**

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| **Operator:** | University of Utah | **Rig:** | Frontier Rig 16 | **Spud Date:** | 28-Jun-21 | **Daily Cost / Mud ($):** | | --- |
| **Measured Depth (ft):** | 421 | **Last Casing:** | 16.000 at 416 | **Wellbore:** | Original Wellbore | **AFE No.** | **AFE ($)** | **Actual ($)** |
| **Vertical Depth (ft):** | | **Next Casing:** | 11.750 at 3,300 | **RKB Elevation (ft):** | 30.40 | --- | --- | --- |
| **Proposed TD (ft):** | 9500 | **Last BOP Test:** | 28-Jun-21 | **Job Reference RKB (ft):** | | --- | --- | --- |
| **Hole Made (ft) / Hrs:** | 293 / 4.0 | **Next BOP Test:** | 19-Jul-21 | **Working Interest:** | | **Totals:** | --- | --- |
| **Average ROP (ft/hr):** | 73.25 | | | | | **Well Cost ($):** | | --- |
| **Drilling Days (act./plan):** | 2/21 | **Flat Days (act./plan):** | 0/9 | **Total Days (act./plan):** | 2/30 | **Days On Location:** | | 2 |
| **Pers/Hrs:** **Operator:** | 3 / 36 | **Contractor:** | 14 / 168 | **Service:** | 4 / 48 | **Other:** | 3 / 36 | **Total:** 24 / 288 |

**Safety Summary:** No incidents or events reported. Conducted BOP Test, Crown Check, Safety Meeting.

**Current Operations:** Testing out choke manifold at report time.

**Planned Operations:**
Test casing along with BOPE.
Pick up 8” drill collars.
Trip in the hole with tri-cone bit.
Drill out shoe track and 10 ft of new formation to perform LOT
Pull out of the hole and lay down tricone bit
Make up 14 3/4” BHA with PDC and continue drilling 14 3/4" section.

**Toolpusher:** Steve Caldwell , Justin Bristol

**Wellsite Supervisors:** Virgil Welch , Brian Gresham **Tel No.:** 7132807438

## Operations Summary

| From | To | Elapsed | End MD(ft) | Code | Operations Description | Non-Prod |
|---|---|---|---|---|---|---|
| 0:00 | 1:00 | 1.00 | 128 | 3-34-3 | Make up 22" surface hole assembly. | |
| 1:00 | 1:30 | 0.50 | 128 | 03-051 | Filled pipe and hole. | |
| 1:30 | 5:00 | 3.50 | 421 | 03-021 | Drilled 22" hole from 128' to 421'. | |
| 5:00 | 5:30 | 0.50 | 421 | 03-051 | Circulated well bore clean. | |
| 5:30 | 6:00 | 0.50 | 421 | 3-27 | Wiped hole f/ 421' t/ 100' with no issues. No fill on bottom. | |
| 6:00 | 6:30 | 0.50 | 421 | 3-6-2 | Pull out of the hole f/ 421' | |
| 6:30 | 7:00 | 0.50 | 421 | 3-34-4 | Lay down NMDC and bit | |
| 7:00 | 8:00 | 1.00 | 421 | 4-56 | Rig up casing crew. | |
| 8:00 | 10:00 | 2.00 | 421 | 4-12-1 | Rigged up and Ran 9 joints of 16" k-55 64# BTC casing (382.89') of total casing. Landed out with 33' landing joint. Shoe set at 416' KB. Float collar at 372' KB. | |
| 10:00 | 11:00 | 1.00 | 421 | 4-56 | Rig down casing crew | |
| 11:00 | 12:30 | 1.50 | 421 | 4-12-1 | Land casing with well head 3' below ground level. | |
| 12:30 | 13:00 | 0.50 | 421 | 5-56 | RIH with stab-in and stab into float collar at 378 | |
| 13:00 | 14:30 | 1.50 | 421 | 5-56 | Rig up cement crew | |
| 14:30 | 16:00 | 1.50 | 421 | 5-99 | Pressure test to 1500 psi, Pump 20 bbl fresh water, pump 50 bbls of sepiolite, pump 5 bbl of fresh water, pump 13 bbl od sodium silkicate, pump 5 bbls fresh water, Mix and pump 146 bbl of 13.88 RC Theremelite w/ 2% CC, displace w/ 6 bbls fresh water, check floats CIP @ 16:00 | |
| 16:00 | 16:30 | 0.50 | 421 | 5-56 | Rig down cement crew | |
| 16:30 | 17:00 | 0.50 | 421 | 5-6-2 | POH with stab-in | |
| 17:00 | 0:00 | 7.00 | 421 | 1-14 | Installed BOP as followed. 16” x 20” 2M DSA, 20” 2M spacer spool, 20” 2M x 20” 3M DSA, 20” 3M mud cross, 20” 3M X 20” 2M DSA, 20” 2M double gates, 20” 2M annular, 20” 2M flow spool. | |

## Management Summary

Drilled 22" hole from 128' to 421'. Circulated hole clean. Wiped hole from 421’ to 100’.
Pulled out laying down tools.
Rigged up and ran 9 joints of 65 ppf, K-55 16” Buttress casing setting shoe at 416’ KB.
Ran in the hole with stab-in tool.
Rigged up Resource cementing and cemented casing in place CIP @ 16:00.
Rigged down Resource cementing.
Pulled out of the hole with stab-in tool.
Installed 20” BOPE. Nippled up BOPE.

## Comments

Good cement to surface.

# Daily Drilling Report

**Well ID: Forge 78B-32**

Field: UTAHFORGE

## Geothermal Resource Group, Inc.

**Well Name: 78B-32**

Sect: 32 Town: 26 Rng: 9W County: Beaver State: UT

**Report No:** 3 **Report For 29-Jun-21**

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| **Operator:** | University of Utah | **Rig:** | Frontier Rig 16 | **Spud Date:** | 28-Jun-21 | **Daily Cost / Mud ($):** | | --- |
| **Measured Depth (ft):** | 433 | **Last Casing:** | 16.000 at 416 | **Wellbore:** | Original Wellbore | **AFE No.** | **AFE ($)** | **Actual ($)** |
| **Vertical Depth (ft):** | 433 | **Next Casing:** | 11.750 at 3,300 | **RKB Elevation (ft):** | 30.40 | --- | --- | --- |
| **Proposed TD (ft):** | 9500 | **Last BOP Test:** | 29-Jun-21 | **Job Reference RKB (ft):** | | --- | --- | --- |
| **Hole Made (ft) / Hrs:** | 12 / 1.5 | **Next BOP Test:** | 19-Jul-21 | **Working Interest:** | | **Totals:** | --- | --- |
| **Average ROP (ft/hr):** | 8.0 | | | **LOT (lbs/gal):** | 19.20 | **Well Cost ($):** | | --- |
| **Drilling Days (act./plan):** | 3/21 | **Flat Days (act./plan):** | 0/9 | **Total Days (act./plan):** | 3/30 | **Days On Location:** | | 3 |

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| **Pers/Hrs:** | **Operator:** | 3 / 36 | **Contractor:** | 14 / 168 | **Service:** | 4 / 48 | **Other:** | 3 / 36 | **Total:** 24 / 288 |

**Safety Summary:** No incidents or events reported. Conducted BOP Test, Crown Check, Safety Meeting.

**Current Operations:** Picked up RSS and ran in the hole. Drilled 14 3/4" hole section from 433' to 1,524' (report time)

**Planned Operations:** Drill 14 3/4" section.

**Toolpusher:** Steve Caldwell , Justin Bristol

**Wellsite Supervisors:** Virgil Welch , Brian Gresham **Tel No.:** 7132807438

## Operations Summary

| From | To | Elapsed | End MD(ft) | Code | Operations Description | Non-Prod |
|---|---|---|---|---|---|---|
| 0:00 | 3:00 | 3.00 | 421 | 3-15 | Installed inner and outer choke and kill valves. Installed choke and kill lines. Torqued all flanges. | |
| 3:00 | 6:00 | 3.00 | 421 | 3-15 | Attempted to Test BOPE system to 300 psi low to 1,000 psi high. First Low test of 300 psi testing against the pipe rams held. Stepped pressure up to 1,000 psi and pressure bleed back down to 500 psi. Attempted to pressure back up to 1,000 psi, pressure bleed back down to 500 psi. Inspected all surface equipment for leaks. No leaks were found. Decision was made to test choke manifold while allowing more time for cement to set. Tested out all valves on choke manifold to 300 psi low for 5 mins and 1,000 psi high for 10 mins. | |
| 6:00 | 9:00 | 3.00 | 421 | 2-14 | Nipple up Flow nipple w/ rotating head to flow line | |
| 9:00 | 10:00 | 1.00 | 421 | 2-14 | Walk catwalk into place and raise V-Door | |
| 10:00 | 11:30 | 1.50 | 421 | 3-34-3 | Set racks and move and strap 15 each 8" DCS | |
| 11:30 | 14:30 | 3.00 | 421 | 3-34-3 | Pick up 15 each 8" DCS and stand back in Derrick | |
| 14:30 | 16:30 | 2.00 | 421 | 3-15 | Test Blind rams to 250 psi for 5 min low & 1000 psi for 10 min high / test good / PU stand of drill pipe and test pipe rams to 250 psi for 5 min & 1000 psi for 10 min test good, test bag to 250 psi for 5 min low & 1000 psi high for 10 min | |
| 16:30 | 18:00 | 1.50 | 421 | 3-34-3 | PU 14-3/4" bit, bit sub, 3 stands of DCS and 1 stand of HWDP and RIH to drill float, cement and shoe | |
| 18:00 | 18:30 | 0.50 | 421 | 2-14 | Installed rotating rubber bushing to bearing assembly. Installed rotating rubber. | |
| 18:30 | 19:30 | 1.00 | 421 | 3-28 | Drilled out shoe track from 370' to 416'. Good cement Throughout shoe track. | |
| 19:30 | 20:00 | 0.50 | 421 | SERV | Serviced rig. | |
| 20:00 | 20:45 | 0.75 | 421 | 3-28 | Cleaned out rat hole to 421'. Drilled new formation to 433' | |
| 20:45 | 21:00 | 0.25 | 421 | 3-5-1 | Circulated hole clean for FIT. | |
| 21:00 | 22:00 | 1.00 | 421 | 3-46 | Performed FIT. | |
| 22:00 | 23:30 | 1.50 | 421 | 3-6-2 | Pulled out of the hole from 433' to surface. Broke out bit along with bit sub. | |
| 23:30 | 0:00 | 0.50 | 421 | 3-98 | Held detailed safety meeting with all involved personal on picking up BHA. | |

## Management Summary

Installed inner and outer choke and kill valves.
Torqued up all breaks on BOPE.
Attempted to test BOPE. High pressure test leaked off. Decision was made to allow more time for cement to set.
Tested out choke manifold.
Nippled up riser and installed flowline.
Walked catwalk into place and raised V-Door.
Set racks in place. Strapped 15-8” drill collars. Picked up 5 stands of 8” drill collars and racked back in the derrick.
Tested out BOPE. Good test.
Made up 14 3/4” slick assembly.
Tripped in the hole.
Drilled out shoe track from 370’ to 416’.
Drilled new formation to 433’.
Circulated hole clean.
Performed FIT to a 1 psi/ft. Good test.
Pulled out of the hole.
Held detailed safety meeting with all involved personal on picking up BHA.